

Table 615. ETF register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFD0	ETF_PIDR4	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CON[3:0]			
	Reset value																									0	0	0	0	0	1	0	0
0xFD4 to 0xFDC	Reserved	Reserved																															
0xFE0	ETF_PIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
	Reset value																									0	1	1	0	0	0	0	1
0xFE4	ETF_PIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
	Reset value																									1	0	1	1	1	0	0	1
0xFE8	ETF_PIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC		JEP106ID[6:4]	
	Reset value																									0	0	0	1	1	0	1	1
0xFEC	ETF_PIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
	Reset value																									0	0	0	0	0	0	0	0
0xFF0	ETF_CIDR0	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
	Reset value																									0	0	0	0	1	1	0	1
0xFF4	ETF_CIDR1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
	Reset value																									1	0	0	1	0	0	0	0
0xFF8	ETF_CIDR2	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
	Reset value																									0	0	0	0	0	1	0	1
0xFFC	ETF_CIDR3	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
	Reset value																									1	0	1	1	0	0	0	1

60.5.6 Trace port interface unit (TPIU)

The TPIU is a CoreSight™ component that formats the trace stream and outputs it on the external trace port signals. The TPIU has a single ATB slave port for incoming trace data. The trace port is a synchronous parallel port, comprising a clock output, TRACECK, and four data outputs, TRACED(7:0). The trace port width is programmable in the range 1 to 8. Using a smaller port width reduces the number of test points/connector pins needed, and frees up I/Os for other purposes. However it restricts the bandwidth of the trace port and hence the quantity of trace information that can be output in real time. **The TRACECK output must be enabled by setting the TRACECLKEN bit in the DBGMCU_CR register before trace is sent to the TPIU. Furthermore, the TRACECK frequency can be programmed in the RCC.**

For more information on the Trace port interface CoreSight™ component, refer to the Arm® CoreSight™ SoC-400 technical reference manual [2].

TPIU registers**TPIU supported port size register (TPIU_SUPPSIZE)**

Address offset: 0x000

Reset value: 0x0000 000F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PORTSIZE[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PORTSIZE[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **PORTSIZE[31:0]**: Indicates supported trace port sizes, from 1 to 32 pins. Bit n-1 when set indicates that port size n is supported.

0x0000 000F: Port sizes 1 to 4 supported

TPIU current port size register (TPIU_CURPSIZE)

Address offset: 0x004

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PORTSIZE[31:16]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PORTSIZE[15:0]															
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:0 **PORTSIZE[31:0]**: Current trace port size

Setting bit n-1 indicates that the current port size is n-pin wide. The value of n must be within the range of supported port sizes (1-4). Only one bit can be set, or unpredictable behavior may result. This register should only be modified when the formatter is stopped.

TPIU supported trigger modes register (TPIU_SUPTRGM)

Address offset: 0x100

Reset value: 0x0000 011F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TRGRUN	TRGD
														r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	TCOUNT 8	Res.	Res.	Res.	MULT 64K	MULT 256	MULT 16	MULT4	MULT2
							r				r	r	r	r	r

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **TRGRUN**: Trigger running

0: Trigger has not occurred or counter is at 0

1: Trigger has occurred and counter is not at 0

Bit 16 **TRIGD**: Triggered

0: Trigger has not occurred

1: Trigger has occurred and counter has reached 0

Bits 15:9 Reserved, must be kept at reset value.

Bit 8 **TCOUNT8**: 8-bit counter register

1: Implemented

Bits 7:5 Reserved, must be kept at reset value.

Bit 4 **MULT64K**: Multiplying the trigger counter by 65536 support

1: Supported

Bit 3 **MULT256**: Multiplying the trigger counter by 256 support

1: Supported

Bit 2 **MULT16**: Multiplying the trigger counter by 16 support

1: Supported

Bit 1 **MULT4**: Multiplying the trigger counter by 4 support

1: Supported

Bit 0 **MULT2**: Multiplying the trigger counter by 2 support

1: Supported

TPIU trigger counter value register (TPIU_TRGCNT)

Address offset: 0x104

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TRIGCOUNT[7:0]							
								r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **TRIGCOUNT[7:0]**: Enable trigger delay indication

Enables delaying trigger flagging to any external connected trace capture or storage devices. This counter is only eight bits wide and is intended to be used only with the counter multipliers in the Trigger multiplier register, 0x108. When a trigger is started, this value, in conjunction with the multiplier, specifies the number of words before the trigger is indicated. When the trigger counter reaches 0, the value written here is reloaded. Writing to this register causes the trigger counter value to reset but does not reset any value on the multiplier. Reading this register returns the preset value, not the current count.

TPIU trigger multiplier register (TPIU_TRGMULT)

Address offset: 0x108

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MULT64K	MULT256	MULT16	MULT4	MULT2
											rw	rw	rw	rw	rw

Bits 31:5 Reserved, must be kept at reset value.

Bit 4 **MULT64K**: Multiply the trigger counter by 65536

0: Disabled

1: Enabled

Bit 3 **MULT256**: Multiply the trigger counter by 256

0: Disabled

1: Enabled

Bit 2 **MULT16**: Multiply the trigger counter by 16

0: Disabled

1: Enabled

Bit 1 **MULT4**: Multiply the trigger counter by 4

0: Disabled

1: Enabled

Bit 0 **MULT2**: Multiply the trigger counter by 2

0: Disabled

1: Enabled

TPIU supported test patterns/modes register (TPIU_SUPTPM)

Address offset: 0x200

Reset value: 0x0003 000F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCONT EN	PTIME EN
														r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PATF0	PATA5	PATW0	PATW1
												r	r	r	r

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **PCONTEN**: Continuous mode support
1: Supported

Bit 16 **PTIMEEN**: Timed mode support
1: Supported

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **PATF0**: Support of FF/00 pattern
Indicates whether the FF/00 pattern is supported as output over the trace port.
1: Supported

Bit 2 **PATA5**: Support of AA/55 pattern
Indicates whether the AA/55 pattern is supported as output over the trace port.
1: Supported

Bit 1 **PATW0**: Support of walking 0's pattern
Indicates whether the walking 0's pattern is supported as output over the trace port.
1: Supported

Bit 0 **PATW1**: Support of walking 1's pattern
Indicates whether the walking 1's pattern is supported as output over the trace port.
1: Supported

TPIU current test pattern/mode register (TPIU_CURTPM)

Address offset: 0x204

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCONT EN	PTIME EN
														rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PATF0	PATA5	PATW0	PATW1
												rw	rw	rw	rw

Bits 31:18 Reserved, must be kept at reset value.

Bit 17 **PCONTEN**: Continuous mode enable
0: Disabled
1: Enabled

Bit 16 **PTIMEEN**: Timed mode enable
0: Disabled
1: Enabled

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **PATF0**: FF/00 pattern enable
Indicates whether the FF/00 pattern is enabled as output over the trace port
0: Disabled
1: Enabled

Bit 2 **PATA5**: AA/55 pattern is enable

Indicates whether the AA/55 pattern is enabled as output over the trace port

0: Disabled

1: Enabled

Bit 1 **PATW0**: Walking 0's pattern enable

Indicates whether the walking 0's pattern is enabled as output over the trace port

0: Disabled

1: Enabled

Bit 0 **PATW1**: Walking 1's pattern enable

Indicates whether the walking 1's pattern is enabled as output over the trace port

0: Disabled

1: Enabled

TPIU test pattern repeat counter register (TPIU_TPRCR)

Address offset: 0x208

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PATTCOUNT[7:0]							
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PATTCOUNT[7:0]**: Number of TRACECLKIN cycles

The field provides a 8-bit counter value to indicate the number of TRACECLKIN cycles for which a pattern runs before it switches to the next pattern.

TPIU formatter and flush status register (TPIU_FFSR)

Address offset: 0x300

Reset value: 0x0000 0002

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TCPRE SENT	FTSTO PPED	FLINPR OG
													r	r	r

Bits 31:3 Reserved, must be kept at reset value.

Bit 2 **TCPRESENT**: TRACECTL output pin availability

Indicates whether the optional TRACECTL output pin is available for use.

0: TRACECTL pin is not present in this device.

Bit 1 **FTSTOPPED**: Formatter stopped

The formatter has received a stop request signal and all trace data and post-amble is sent. Any additional trace data on the ATB interface is ignored.

0: Formatter has not stopped

1: Formatter has stopped

Bit 0 **FLINPROG**: Flush in progress

Indicates whether a flush on the ATB slave port is in progress. This bit reflects the status of the AFVALIDS output. A flush can be initiated by the flush control bits in the TPIU_FFCR register.

0: No flush in progress

1: Flush in progress

TPIU formatter and flush control register (TPIU_FFCR)

Address offset: 0x304

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	STOPT RIG	STOPF L	Res.	TRIGF L	TRIGE VT	TRIGIN	Res.	FONMA N	FONTR IG	FONFLI N	Res.	Res.	ENFCO NT	EN FTC
		rw	rw		rw	rw	rw		rw	rw	rw			rw	rw

Bits 31:14 Reserved, must be kept at reset value.

Bit 13 **STOPTRIG**: Stop on trigger event

0: No effect

1: Stop formatter when a trigger event occurs

Bit 12 **STOPFL**: Stop on flush

0: No effect

1: Stop formatter when flush is completed

Bit 11 Reserved, must be kept at reset value.

Bit 10 **TRIGFL**: Trigger on flush

0: No effect

1: Indicate a trigger in the trace stream when flush is completed

Bit 9 **TRIG EVT**: Trigger on trigger event

0: No effect

1: Indicate a trigger in the trace stream when trigger event occurs

Bit 8 **TRIGIN**: Trigger on trigger in

0: No effect

1: Indicate a trigger in the trace stream when the TRIGIN input from the system CTI is asserted.

Bit 7 Reserved, must be kept at reset value.

Bit 6 **FONMAN**: Generate a manual flush

0: No effect

1: Flush the trace

This bit is cleared automatically when the flush completes.

Bit 5 **FONTRIG**: Flush on trigger event

A trigger event occurs when the trigger counter reaches 0, or, if the trigger counter is 0, when the TRIGIN input from the system CTI is high.

0: No effect

1: Flush the trace if a trigger event occurs

Bit 4 **FONFLIN**: Flush on flush in

0: No effect

1: Flush the trace if the FLUSHIN input from the system CTI is asserted

Bits 3:2 Reserved, must be kept at reset value.

Bit 1 **ENFCONT**: Enable continuous formatting

0: Continuous formatting is disabled

1: Continuous formatting is enabled

Bit 0 **ENFTC**: Enable the embedding of triggers in formatted trace

0: Formatting is disabled

1: Formatting is enabled

TPIU formatter synchronization counter register (TPIU_FSCR)

Address offset: 0x400

Reset value: 0x0000 0040

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	CYCCOUNT[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **CYCCOUNT**[11:0]: Enables effective use of TPAs

Enables effective use of different-sized TPAs without wasting large amounts of storage capacity of the capture device. This counter contains the number of formatter frames since the last synchronization packet of 128 bits. It is a 12-bit counter with a maximum count value of 4096. This equates to synchronization every 65536 bytes, that is, 4096 packets x 16 bytes per packet. The default is set up for a synchronization packet every 1024 bytes, that is, every 64 formatter frames. If the formatter is configured for continuous mode, full and half-word sync frames are inserted during normal operation. Under these circumstances, the count value is the maximum number of complete frames between full synchronization packets.

TPIU claim tag set register (TPIU_CLAIMSET)

Address offset: 0xFA0

Reset value: 0x0000 000F

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMSET[3:0]			
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **CLAIMSET[3:0]**: Set claim tag bits

Write:

0000: No effect

xxx1: Set bit 0

xx1x: Set bit 1

x1xx: Set bit 2

1xx: Set bit 3

Read:

0xF: Indicates there are four bits in claim tag

TPIU claim tag clear register (TPIU_CLAIMCLR)

Address offset: 0xFA4

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLAIMCLR[3:0]			
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value.

Bits 3:0 **CLAIMCLR[3:0]**: Reset claim tag bits

Write:

0000: No effect

xxx1: Clear bit 0

xx1x: Clear bit 1

x1xx: Clear bit 2

1xxx: Clear bit 3

Read: Returns current value of claim tag

TPIU lock access register (TPIU_LAR)

Address offset: 0xFB0

Reset value: 0xFFFF XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ACCESS_W[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACCESS_W[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **ACCESS_W[31:0]**: TPIU register access enable

Enables write access to some TPIU registers by processor cores (debuggers do not need to unlock the component)

0xC5ACCE55: Enable write access

Other values: Disable write access

TPIU lock status register (TPIU_LSR)

Address offset: 0xFB4

Reset value: 0x0000 0003

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LOCKT YPE	LOCKG RANT	LOCKE XIST
													r	r	r

Bits 31:3 Reserved, must be kept at reset value.

Bit 2 **LOCKTYPE**: Size of the TPIU_LAR register
0: 32-bit

Bit 1 **LOCKGRANT**: Current status of lock
This bit always returns zero when read by an external debugger.

0: Write access is permitted
1: Write access is blocked. Only read access is permitted.

Bit 0 **LOCKEXIST**: Implementation of a lock control mechanism
The bit indicates whether a lock control mechanism is implemented. It always returns zero when read by an external debugger.

0: No lock control mechanism is available
1: Lock control mechanism is implemented

TPIU authentication status register (TPIU_AUTHSTAT)

Address offset: 0xFB8

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SNID[1:0]		SID[1:0]		NSNID[1:0]		NSID[1:0]	
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:6 **SNID[1:0]**: Security level for secure non-invasive debug
0x0: Not implemented

Bits 5:4 **SID[1:0]**: Security level for secure invasive debug
0x0: Not implemented

Bits 3:2 **NSNID[1:0]**: Security level for non-secure non-invasive debug
0x0: Not implemented

Bits 1:0 **NSID[1:0]**: Security level for non-secure invasive debug
0x0: Not implemented

TPIU device configuration register (TPIU_DEVID)

Address offset: 0xFC8

Reset value: 0x0000 00A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	SWO UART NRZ	SWO MAN	TCLK DATA	FIFO SIZE[2:0]			CLK RELAT	MUXNUM[4:0]				
				r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:12 Reserved, must be kept at reset value.

Bit 11 **SWOUARTNRZ**: Support of SWO UART or NRZ

Indicates whether serial wire output, UART or NRZ, is supported.

0: Not supported

Bit 10 **SWOMAN**: Support of SWO Manchester format

Indicates whether serial wire output, Manchester encoded format, is supported.

0: Not supported

Bit 9 **TCLKDATA**: Support of trace clock plus data

0: Not supported

Bits 8:6 **FIFOSIZE[2:0]**: FIFO size in powers of 2

0x2: FIFO size = 4 (16 bytes)

Bit 5 **CLKRELAT**: ATB clock and TRACECLKIN relation

Indicates the relationship between the ATB clock and TRACECLKIN (synchronous or asynchronous)

1: Asynchronous

Bits 4:0 **MAXNUM[4:0]**: Number/type of ATB input port multiplexing

0x0: None

TPIU device type identifier register (TPIU_DEVTYPE)

Address offset: 0xFCC

Reset value: 0x0000 0011

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SUBTYPE[3:0]			MAJORTYPE[3:0]				
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **SUBTYPE[3:0]**: Sub-classification
0x1: Trace port component

Bits 3:0 **MAJORTYPE[3:0]**: Major classification
0x1: Trace sink component

TPIU CoreSight peripheral identity register 4 (TPIU_PIDR4)

Address offset: 0xFD0

Reset value: 0x0000 0004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	4KCOUNT[3:0]				JEP106CON[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **4KCOUNT[3:0]**: Register file size
0x0: Register file occupies a single 4 Kbyte region

Bits 3:0 **JEP106CON[3:0]**: JEP106 continuation code
0x4: Arm® JEDEC code

TPIU CoreSight peripheral identity register 0 (TPIU_PIDR0)

Address offset: 0xFE0

Reset value: 0x0000 0012

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PARTNUM[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PARTNUM[7:0]**: Part number field, bits [7:0]
0x12: TPIU part number

TPIU CoreSight peripheral identity register 1 (TPIU_PIDR1)

Address offset: 0xFE4

Reset value: 0x0000 00B9

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	JEP106ID[3:0]				PARTNUM[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **JEP106ID[3:0]**: JEP106 identity code field, bits [3:0]

0xB: Arm® JEDEC code

Bits 3:0 **PARTNUM[11:8]**: Part number field, bits [11:8]

0x9: TPIU part number

TPIU CoreSight peripheral identity register 2 (TPIU_PIDR2)

Address offset: 0xFE8

Reset value: 0x0000 005B

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVISION[3:0]				JEDEC	JEP106ID[6:4]		
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVISION[3:0]**: Component revision number

0x5: r1p0

Bit 3 **JEDEC**: JEDEC assigned value

1: Designer ID specified by JEDEC

Bits 2:0 **JEP106ID[6:4]**: JEP106 identity code field, bits [6:4]

0x3: Arm® JEDEC code

TPIU CoreSight peripheral identity register 3 (TPIU_PIDR3)

Address offset: 0xFEC

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	REVAND[3:0]				CMOD[3:0]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **REVAND[3:0]**: Metal fix version

0x0: No metal fix

Bits 3:0 **CMOD[3:0]**: Customer modified

0x0: No customer modifications

TPIU CoreSight component identity register 0 (TPIU_CIDR0)

Address offset: 0xFF0

Reset value: 0x0000 000D

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[7:0]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[7:0]**: Component ID field, bits [7:0]

0x0D: Common ID value

TPIU CoreSight component identity register 1 (TPIU_CIDR1)

Address offset: 0xFF4

Reset value: 0x0000 0090

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLASS[3:0]				PREAMBLE[11:8]			
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:4 **CLASS[3:0]**: Component ID field, bits [15:12] - component class

0x9: CoreSight component

Bits 3:0 **PREAMBLE[11:8]**: Component ID field, bits [11:8]

0x0: Common ID value

TPIU CoreSight component identity register 2 (TPIU_CIDR2)

Address offset: 0xFF8

Reset value: 0x0000 0005

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[19:12]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[19:12]**: Component ID field, bits [23:16]

0x05: Common ID value

TPIU CoreSight component identity register 3 (TPIU_CIDR3)

Address offset: 0xFFC

Reset value: 0x0000 00B1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREAMBLE[27:20]							
								r	r	r	r	r	r	r	r

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:0 **PREAMBLE[27:20]**: Component ID field, bits [31:24]

0xB1: Common ID value

TPIU register map and reset values**Table 616. TPIU register map and reset values**

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x000	TPIU_SUPPSIZE	PORTSIZE[31:0]																																	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1		
0x004	TPIU_CURPSIZE	PORTSIZE[31:0]																																	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
0x008 to 0x0FC	Reserved	Reserved																																	
0x100	TPIU_SUPTRGM	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TRGRUN	TRIGD	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TCOUNT8	Res.	Res.	Res.	Res.	MULT64K	MULT256	MULT16	MULT4	MULT2	
	Reset value															0	0								1					1	1	1	1	1	
0x104	TPIU_TRGCNT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	TRIGCOUNT[7:0]										
	Reset value																									0	0	0	0	0	0	0	0	0	
0x108	TPIU_TRGMULT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	MULT64K	MULT256	MULT16	MULT4	MULT2		
	Reset value																											0	0	0	0	0	0	0	
0x10C to 0x1FC	Reserved	Reserved																																	
0x200	TPIU_SUPTPM	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCONTEN	PTIMEEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PATF0	PATA5	PATW0	PATW1		
	Reset value															1	1													1	1	1	1		
0x204	TPIU_CURTPM	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PCONTEN	PTIMEEN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PATF0	PATA5	PATW0	PATW1		
	Reset value															0	0													0	0	0	0		

Table 616. TPIU register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0x208	TPIU_TPCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PATT[7:0]													
	Reset value																									0	0	0	0	0	0	0	0						
0x20C to 0x2FC	Reserved	Reserved																																					
0x300	TPIU_FFSR	Res																															TCPRESENT						
	Reset value																															0	1	0					
0x304	TPIU_FFCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ENFCONT	ENFTC	FLINPROG					
	Reset value																			0	0	0	0	0	0		0	0	0	0		0	0						
0x308	TPIU_FSCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res					
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0						
0x30C to 0x3FC	Reserved	Reserved																																					
0xFA0	TPIU_CLAIMSET	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res						
	Reset value																														1	1	1	1					
0xFA4	TPIU_CLAIMCLR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res					
	Reset value																														0	0	0	0					
0xFB0	TPIU_LAR	ACCESS_W[31:0]																																					
	Reset value	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-						
0xFB4	TPIU_LSR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res					
	Reset value																														0	1	1	1					
0xFB8	TPIU_AUTHSTAT	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res					
	Reset value																														0	0	0	0					
0xFBC to 0xFC4	Reserved	Reserved																																					
0xFC8	TPIU_DEVID	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res					
	Reset value																					0	0	0	0	0	1	0	1	0	0	0	0	0					

Table 616. TPIU register map and reset values (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFCC	TPIU_DEVTYPE	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	SUBTYPE[3:0]			MAJORTYP[3:0]				
	Reset value																									0	0	0	1	0	0	0	1
0xFD0	TPIU_PIDR4	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	4KCOUNT[3:0]			JEP106CON[3:0]				
	Reset value																									0	0	0	0	0	1	0	0
0xFD4 to 0xFDC	Reserved	Reserved																															
0xFE0	TPIU_PIDR0	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PARTNUM[7:0]							
	Reset value																									0	0	0	1	0	0	1	0
0xFE4	TPIU_PIDR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	JEP106ID[3:0]			PARTNUM[11:8]				
	Reset value																									1	0	1	1	1	0	0	1
0xFE8	TPIU_PIDR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	REVISION[3:0]			JEDC	JEP106ID[6:4]			
	Reset value																									0	1	0	1	1	1	0	1
0xFEC	TPIU_PIDR3	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	REVAND[3:0]			CMOD[3:0]				
	Reset value																									0	0	0	0	0	0	0	0
0xFF0	TPIU_CIDR0	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PREAMBLE[7:0]							
	Reset value																									0	0	0	0	1	1	0	1
0xFF4	TPIU_CIDR1	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	CLASS[3:0]			PREAMBLE[11:8]				
	Reset value																									1	0	0	1	0	0	0	0
0xFF8	TPIU_CIDR2	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PREAMBLE[19:12]							
	Reset value																									0	0	0	0	0	1	0	1
0xFFC	TPIU_CIDR3	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	PREAMBLE[27:20]							
	Reset value																									1	0	1	1	0	0	0	1

60.5.7 Serial wire output (SWO) and SWO trace funnel (SWTF)

The SWO is a CoreSight component that formats the trace stream from the processor ITM and outputs it on the single wire TRACESWO output. The SWO trace funnel (SWTF) must be programmed to enable the trace bus from the Cortex-M7 ITM before trace is enabled. The SWTF registers are listed in [Table 618](#).

Compared to the TPIU, the SWO contains:

- no formatter
- no pattern generator
- an 8-bit ATB input
- no synchronous trace output, that is, no TRACEDATA or TRACECLK pins
- no support for flush, because this is not required
- no support for triggering

The SWO output supports Manchester encoded and UART NRZ formats.